

Analytic Study of Double Descent in Binary Classification: The Impact of Loss

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Abstract—Extensive empirical evidence reveals that, for a wide range of different learning methods and data sets, the risk curve exhibits a double-descent (DD) trend as a function of the model size. In our recent coauthored paper [Deng et al., '19], we proposed simple binary linear classification models and showed that the test error of gradient descent (GD) with logistic loss undergoes a DD. In this paper, we complement these results by extending them to GD with square loss. We show that the DD phenomenon persists, but we also identify several differences compared to logistic loss. This emphasizes that crucial features of DD curves (such as their transition threshold and global minima) depend both on the training data and on the learning algorithm. We further study the dependence of DD curves on the size of the training set. Similar to [Deng et al., '19] our results are analytic: we plot the DD curves by first deriving sharp asymptotics for the test error under Gaussian features. Albeit simple, the models permit a principled study, the outcomes of which theoretically corroborate related empirical findings occurring in more complex learning tasks.

I. INTRODUCTION

Motivation. It is common practice in modern learning architectures to choose the number of training parameters such that it overly exceeds the size of the training set. Despite the *overparametrization*, such architectures are known to generalize well in practice. Specifically, it has been consistently reported in the recent literature, that the risk curve as a function of the model size exhibits a, so called, “double-descent” (DD) [1]. The initial descent is followed by an ascend as predicted by conventional statistical wisdom of the approximation-generalization tradeoff, but as the model size surpasses a certain transition threshold a second descent occurs. This phenomenon has been demonstrated experimentally for decision trees, random features and two-layer neural networks (NN) in [1] and, more recently, for deep NNs (including ResNets, standard CNNs and Transformers) in [2]; see also [3], [4] for similar observations. Our work is motivated by these empirical findings and has a two-fold goal. First, we demonstrate analytically that the DD phenomenon is already present in certain simple binary linear classification settings. Second, for these simple settings, we undertake a principled study of how certain features of DD curves vary depending on the data as well as on the training algorithms. The outcomes of this study theoretically corroborate related empirical findings occurring in more complicated learning methods for a variety of learning tasks. Different to our work, most other recent efforts towards theoretically understanding DD curves focus on linear regression.

Contributions. First, we evaluate the classification error of gradient descent (GD) with least-squares (LS) loss for two simple, yet popular, models for binary linear classification, namely logistic and Gaussian mixtures (GM) models. Specifically, we study a high-dimensional regime in which the size n of training set and the size p of the learnt model both grow large at a proportional rate κ . We derive asymptotic formulae for all values of $\kappa > 0$. Second, we use the theoretical predictions to show that the test error of certain simple classification models undergoes a DD behavior when performing GD on square-loss. The DD curves are qualitatively different than corresponding curves for logistic loss that we studied in our recent coauthored paper [5]. Therefore, we show that the characteristics of the DD curve, such as the location and the shape of its kink, depend not only on the data, but also on the training algorithm. Our results further identify SNR and data model regimes where the test error achieved by LS optimized over the model size p is *lower* than the corresponding error for logistic loss. Finally, we use our theoretical results to study the dependence of the DD curves on the size of the training set: because of the inherent W-shape of the curve, there exist (“bad”) choices of the model size such that the test error of GD increases despite larger sample-size n .

Related works. Recent efforts towards theoretically understanding the phenomenon of DD focus on linear regression settings [6]–[13]; see [2, App. C] for an extended discussion. Out of these, the most related to our paper are [7], [9], which use random matrix theory (RMT) to study DD curves of GD for linear regression with square loss in certain simple mis-match models with Gaussian features. Here, we extend these results to classification. To the best of our knowledge, the only previous works that study DD in a classification setting are [14] and our co-authored paper [5], which do so by computing the asymptotics of GD with logistic loss. In this paper, we extend the results of [5] to GD with square loss, thus emphasizing the dependence of DD not only on the data, but also on the training algorithm. On a technical level, our asymptotic analysis fits in the rapidly growing recent literature on sharp performance guarantees of convex optimization based estimators; see [15]–[22] and many references therein. Specifically, we utilize the convex Gaussian min-max theorem (CGMT) [17] and the machinery put forth in [5], [23]–[28], which applies the CGMT to generalized linear models. There are several other works on sharp asymptotics of binary linear classification both for the logistic [27], [29], [30] and the Gaussian-mixture model [31]–[33]. While closely

related, these works differ in terms of motivation, model specification, end-results, and proof techniques.

II. LEARNING MODEL

A. Two data models

Let $\mathbf{x} \in \mathbb{R}^d$ be the feature vector and $y \in \pm 1$ its class label. We study the following popular data models [34, Sec. 3.1].

A generative Logistic model: We model the marginal $p(y = +1|\mathbf{x})$ with the sigmoid function $(1 + e^{-\mathbf{x}^T \boldsymbol{\eta}_0})^{-1}$, where $\boldsymbol{\eta}_0 \in \mathbb{R}^d$ is an unknown weight vector. Also, we assume IID Gaussian feature vectors $\mathbf{x}$. For compactness let $\text{Rad}(p) \in \pm 1$ denote a symmetric Bernoulli distribution with probability p for the value $+1$. Then, under the logistic model we have

$$y \sim \text{Rad}(f(\mathbf{x}^T \boldsymbol{\eta}_0)), \quad \mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d). \quad (1)$$

A discriminative Gaussian mixtures (GM) model: We model the class-conditional densities $p(\mathbf{x}|y)$ with Gaussians. Specifically, each data point belongs to one of two classes with probabilities $\pi_{\pm 1}$ such that $\pi_{+1} + \pi_{-1} = 1$ and

$$y = \pm 1 \Leftrightarrow \mathbf{x} \sim \mathcal{N}(\pm \boldsymbol{\eta}_0, \mathbf{I}_d). \quad (2)$$

B. Training-data: A mismatch model

We assume access to n training points generated according to either (1) or (2). We allow for the possibility that only a subset $\mathcal{S} \subset [d]$ of the total number of features is known during training. Concretely, for each feature vector $\mathbf{x}_i$, $i \in [n]$, the learner only knows a sub-vector $\mathbf{w}_i := \mathbf{x}_i(\mathcal{S}) := \{\mathbf{x}_i(j), j \in \mathcal{S}\}$, for a chosen set $\mathcal{S} \subset [d]$. We denote the size of the *known* feature sub-vector as $p := |\mathcal{S}|$. Onward, we choose $\mathcal{S} = \{1, 2, \dots, p\}$ ¹, i.e., select the features sequentially in the order in which they appear. Clearly, $1 \leq p \leq d$. Overall, the training set $\mathcal{T}(\mathcal{S})$ consists of n data pairs:

$$\begin{aligned} \mathcal{T}(\mathcal{S}) &:= \{(\mathbf{w}_i, y_i), i = 1, \dots, n\}, \\ \mathbf{w}_i &= \mathbf{x}_i(\mathcal{S}) \in \mathbb{R}^p \quad \text{where } (y_i, \mathbf{x}_i) \sim (1) \text{ or } (2). \end{aligned} \quad (3)$$

We let $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n]^T$ denote the *feature matrix*.

C. Classification algorithm

Having access to the training set $\mathcal{T}(\mathcal{S})$, the learner obtains an estimate $\beta_\star \in \mathbb{R}^p$ of the weight vector $\boldsymbol{\eta}_0 \in \mathbb{R}^d$. For a fresh $\mathbf{x}$ (and $\mathbf{w} := \mathbf{x}(\mathcal{S})$), she forms a *linear classifier* and decides a label $\hat{y} = \text{sign}(\mathbf{w}^T \beta_\star)$. The estimate $\beta_\star$ is minimizing the empirical risk

$$\min_{\beta} \hat{\mathcal{R}}_{\text{emp}}(\beta) := \frac{1}{n} \sum_{i=1}^n \ell(y_i \mathbf{w}_i^T \beta), \quad (4)$$

for certain loss function $\ell : \mathbb{R} \rightarrow \mathbb{R}$. A traditional way to minimize $\hat{\mathcal{R}}_{\text{emp}}$ is by constructing gradient descent (GD) iterates β_k , $k \geq 0$ via $\beta_{(k+1)} = \beta_{(k)} - \eta_k \nabla \hat{\mathcal{R}}_{\text{emp}}(\beta_{(k)})$, for step-size $\eta_k > 0$ and arbitrary $\beta_{(0)}$. We run GD until convergence and set $\beta_\star := \lim_{k \rightarrow \infty} \beta_{(k)}$. We focus on the following two popular choices for the loss function ℓ :

¹This assumption is without loss of generality for our asymptotic model specified in Section III-A.

- Logistic: $\ell(t) := \log(1 + e^{-t})$,
- Least-squares: $\ell(t) := (1 - t)^2$.

For the square loss, note that $(y_i \mathbf{w}_i^T \beta - 1)^2 = (y_i - \mathbf{w}_i^T \beta)^2$ since $y_i \in \{\pm 1\}$.

For a new sample $(\mathbf{x}, y)$ we measure test error of $\hat{y}$ by the expected *risk*

$$\mathcal{R}(\beta_\star) := \mathbb{E}[\mathbb{1}(\hat{y} \neq y)]. \quad (5)$$

The expectation is over the distribution of the *new* data sample $(\mathbf{x}, y)$ generated according to either (1) or (2). In particular, note that $\mathcal{R}(\beta_\star)$ is a function of the training set $\mathcal{T}$. The training error of $\beta_\star$ is given by $\mathcal{R}_{\text{train}}(\beta_\star) := \frac{1}{n} \sum_{i=1}^n \mathbb{1}(\hat{y} \neq y_i)$.

D. Optimization regimes of Gradient Descent

Our analysis of the classification performance of Gradient Descent (GD) makes use of the following characterizations of the converging behavior of GD iterations for a given training set $\mathcal{T}$ and loss function ℓ (logistic and LS).

Logistic loss: For logistic loss the convergence depends critically on the event that the training data are linearly separable, i.e., $\mathcal{E}_{\text{sep}} := \{ \exists \beta \in \mathbb{R}^p : y_i(\mathbf{w}_i^T \beta) \geq 1, \forall i \in [n] \}$. When $\mathcal{E}_{\text{sep}}$ does *not* hold, then $\hat{\mathcal{R}}_{\text{emp}}(\beta)$ is coercive, its sub-level sets are closed and GD iterates converge to a finite minimizer $\hat{\beta}$ of the empirical loss in (4). Furthermore, when the feature matrix is full column rank, then the loss is strongly convex and $\hat{\beta}$ is unique. But, when data are separable, the set of minimizers of the empirical loss is unbounded and [35] shows that the normalized iterations converge to the max-margin classifier, i.e., $\|\frac{\beta^{(k)}}{\|\beta^{(k)}\|_2} - \frac{\hat{\beta}}{\|\hat{\beta}\|_2}\|_2 \xrightarrow{k \rightarrow \infty} 0$, where

$$\hat{\beta} = \arg \min_{\beta} \|\beta\|_2 \quad \text{sub. to } y_i \mathbf{w}_i^T \beta \geq 1, \forall i. \quad (6)$$

Least-Squares: For least-squares loss, it is a well-known fact that for appropriately small and constant step size the GD iterations converge to the min-norm least-squares regression solution (e.g., see [7, Prop. 1]): $\hat{\beta} = \arg \min \{\|\beta\|_2 \mid \beta \text{ minimizes } \|\mathbf{y} - \mathbf{W}\beta\|_2\}$. Specifically, when the feature matrix $\mathbf{W}$ is full column-rank, then the empirical loss is strongly convex and GD converges to $\hat{\beta} = (\mathbf{W}^T \mathbf{W})^{-1} \mathbf{W}^T \mathbf{y}$. In contrast, when $\mathbf{W}$ is full row-rank, then out of all the solutions β that linearly interpolate the data, i.e., $\mathbf{y} = \mathbf{W}\beta$, GD converges to the min-norm solution:

$$\hat{\beta} = \arg \min_{\beta} \|\beta\|_2 \quad \text{sub. to } y_i = \mathbf{w}_i^T \beta, \forall i. \quad (7)$$

III. SHARP ASYMPTOTICS

We present sharp asymptotic formulae for the test error of GD iterations for logistic and square losses under the learning model of Sections II-A and II-B, as well as, the linear asymptotic setting in Section III-A. The first key step of our analysis, makes use of the convergence results discussed in Section II-D that relate GD to the convex programs (4), (6) and (7) depending on whether certain deterministic properties of the training data $\mathcal{T}$ (such as linear separability or conditioning of the feature matrix) hold. In Section III-B, we show that in

the linear asymptotic regime with Gaussian features the change in behavior of convergence of GD undergoes sharp phase-transitions for both loss functions and data models (logistic and Gaussian-mixtures). The boundary of the phase-transition separates the so-called under- and over-parametrized regimes. Thus, for both logistic and LS loss and each one of its two corresponding regimes, GD converges to an associated convex program. The second key-step of our analysis, uses the CGMT [17], [19] to evaluate the test error of these convex programs. This paper focuses on LS. Corresponding results for logistic loss are derived in the coauthored work [5]. In Section IV, we combine the results from both papers to compare DD curves of logistic and LS loss. Due to space constraints, we only present the asymptotic formulae for the case of LS; please see [5, Sec. 3.3 & 3.4] for the case of logistic loss.

A. Asymptotic setting

Our asymptotic results hold in a *linear asymptotic regime* where $n, d, p \rightarrow +\infty$ such that

$$d/n \rightarrow \zeta > 0 \quad \text{and} \quad p/n \rightarrow \kappa \in (0, \zeta], \quad (8)$$

where recall the following: d is the dimension of the ambient space; n is the training sample size; and p is the model size.

To quantify the effect of the overparametrization ratio κ on the test error we let $\beta_0 \in \mathbb{R}^p$ (resp., $\gamma_0 \in \mathbb{R}^{d-p}$) denote the vector of weight coefficients corresponding to the known (resp., unknown) features such that $\eta_0 := [\beta_0^T, \gamma_0^T]^T$. In this notation, we study a sequence of problems of increasing dimensions (n, d, p) as in (8) that further satisfy:

$$\|\eta_0\|_{2 \rightarrow r} \rightarrow r \quad \text{and} \quad \|\beta_0\|_{2 \rightarrow s} := s(\kappa). \quad (9)$$

The parameter $s \in (0, r]$ can be thought of as the useful signal strength. Our notation specifies that $s(\kappa)$ is a function of κ . We are interested in functions $s(\kappa)$ that are increasing in κ such that the signal strength increases as more features enter into the training model; see Sec. IV-A. For each triplet (n, d, p) in the sequence of problems considered, the corresponding training set $\mathcal{T} = \mathcal{T}(\{1, 2, \dots, p\})$ follows (3).

B. Phase-transitions of optimization regimes

As discussed in Section II, the behavior of GD with LS loss changes depending on whether the feature matrix $\mathbf{W}$ is full column- or row-rank. On the one hand, when $p > n$ (aka $\kappa > 1$) and $\mathbf{W}\mathbf{W}^T$ is invertible, then GD converges to the unique LS solution. On the other hand, when $n > p$ (aka $\kappa < 1$) and $\mathbf{W}^T\mathbf{W}$ is invertible, then the data can be linearly interpolated and GD converges to the min-norm solution in (7). Under the Gaussian feature model, the following well-known result shows that the aforementioned change in behavior undergoes a sharp phase-transition with transition threshold 1.

Proposition III.1 (Linear interpolation threshold (e.g. [36])). *For a training set $\mathcal{T}$ that is generated by either of the two models in (1) or (2), the following hold:*

$$\begin{aligned} \kappa > 1 &\Rightarrow \lim_{n, p, d \rightarrow \infty} \Pr(\mathbf{W}\mathbf{W}^T \text{ is invertible}) = 1, \\ \kappa < 1 &\Rightarrow \lim_{n, p, d \rightarrow \infty} \Pr(\mathbf{W}^T\mathbf{W} \text{ is invertible}) = 1. \end{aligned}$$

Put in words: when $\kappa > 1$, with probability 1, the LS empirical loss can be driven to zero and GD converges to the min-norm solution $\hat{\beta}$ of (7). Moreover, since $\forall i : \mathbf{y}_i \mathbf{w}_i^T \hat{\beta} = 1$, the training error is also going to zero in the regime $\kappa > 1$.

Remark 1 (Comparison to Logistic loss). In [5, Prop. 3.1], we prove a corresponding phase-transition for logistic loss. Specifically, for both the Logistic and GM models we compute thresholds $\kappa_{*, \text{LOG}}$ and $\kappa_{*, \text{GM}}$ in $(0, 1/2)$, such that GD with logistic loss converges to: (i) the unique solution of (4) for κ less than the threshold; (ii) to the max-margin solution (6) for κ larger than the threshold. In the latter case, the training data are separable; hence, both the empirical logistic loss and the training error are zero with probability one. According to Proposition III.1, the corresponding threshold for LS loss is 1 for both the logistic and GM models. Note that this is strictly larger (in fact, at least double) than both $\kappa_{*, \text{LOG}}$ and $\kappa_{*, \text{GM}}$. Moreover, both $\kappa_{*, \text{LOG}}$ and $\kappa_{*, \text{GM}}$ are functions of the SNR r , which is *not* the case for LS loss.

C. High-dimensional asymptotics

For each increasing triplet (n, d, p) and sequence of problem instances following the asymptotic setting of Section III-A, let $\hat{\beta} \in \mathbb{R}^d$ be the sequence of vectors of increasing dimension corresponding to the converging point of GD for LS. The Propositions III.2 and III.3 characterize the asymptotic value of the classification error of the sequence of $\hat{\beta}$'s for the logistic and Gaussian-mixture (GM) models, respectively.

In what follows, for a sequence of random variables $\mathcal{X}_{n,p,d}$ that converges in probability to a constant c in the limit of (8), we simply write $\mathcal{X}_{n,p,d} \xrightarrow{P} c$. Furthermore, let $Q(t) = \frac{1}{\sqrt{2\pi}} \int_t^{+\infty} e^{-t^2/2} dt$ be the Gaussian Q-function.

Proposition III.2 (Logistic model). *Fix total signal strength r , over-parametrization ratio $\kappa > 0$ and parameter $s = s(\kappa) \in (0, r]$. With these, consider a training set $\mathcal{T}$ that is generated by the Logistic model in (1) and satisfies (9). Define the following random variables:*

$$H, G, Z \stackrel{iid}{\sim} \mathcal{N}(0, 1), \quad Y \sim \text{Rad}(f(sG + \sqrt{r^2 - s^2}Z)),$$

and denote, $\nu := 2\mathbb{E}_H[\frac{\exp(-rH)}{(1+\exp(-rH))^2}]$. Finally, define the effective noise parameter $\rho := \alpha/\mu$, where μ and $\alpha > 0$ are computed as follows:

$$\mu = \begin{cases} s\nu, & \text{if } \kappa < 1, \\ s\nu/\kappa, & \text{if } \kappa > 1. \end{cases}, \quad \alpha^2 = \begin{cases} (1 - s^2\nu^2)\frac{\kappa}{1-\kappa}, & \text{if } \kappa < 1, \\ \frac{\kappa^2 + (1-2\kappa)s^2\nu^2}{\kappa^2(\kappa-1)}, & \text{if } \kappa > 1. \end{cases} \quad (10)$$

Then, $\mathcal{R}(\hat{\beta}) \xrightarrow{P} \mathbb{P}(\rho H + GY < 0)$.

Proposition III.3 (GM model). *Fix total signal strength r , over-parametrization ratio $\kappa > 0$ and parameter $s = s(\kappa) \in (0, r]$. With these, consider a training set $\mathcal{T}$ that is generated by the GM model in (2) and satisfies (9). Define the effective*

noise parameter $\rho := \alpha/\mu$, where μ and $\alpha > 0$ are computed as follows:

$$\mu = \begin{cases} \frac{s}{1+s^2}, & \text{if } \kappa < 1, \\ \frac{s}{\kappa+s^2}, & \text{if } \kappa > 1, \end{cases}, \quad \alpha^2 := \begin{cases} \frac{\kappa}{1-\kappa} \frac{1}{1+s^2}, & \text{if } \kappa < 1, \\ \frac{\kappa}{(\kappa-1)(\kappa+s^2)^2}, & \text{if } \kappa > 1. \end{cases} \quad (11)$$

Then, $\mathcal{R}(\hat{\beta}) \xrightarrow{P} Q\left(\frac{s}{\sqrt{1+\rho^2}}\right)$.

To prove Propositions III.2 and III.3, we use the fact that (cf. Sec. III-B): (i) for $\kappa < 1$, $\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{W}\beta\|_2$, where the minimizer is unique; (ii) for $\kappa > 1$, $\hat{\beta}$ is as in (7). Thus, it suffices to evaluate the asymptotics of these two convex programs, which we do by using the CGMT². The machinery follows closely (in fact, is simpler) the proofs for logistic loss that we present in [5], wherein Corollary A.1 provides a formal presentation. To the best of our knowledge, the formulae presented in Propositions III.2 and III.3 are novel, with the exception of the result of Proposition III.2 for $\kappa < 1$, which follows from [23], [26].

Remark 2 (The transition threshold). For both the logistic and GM models it can be checked that as κ approaches the linear interpolation threshold 1 from either left or right: $\rho \rightarrow +\infty$. Thus, irrespective of r and s , as $\kappa \rightarrow 1$ (i.e., the model size p approaches the training size n), the test error $\mathcal{R}$ converges to $1/2$ and there is no predictive ability for classification.

IV. NUMERICAL RESULTS & DISCUSSION

A. Feature selection models

The results and discussions that follow are for the following two feature selection models; see also [5], [7], [9], [37].

Linear model: The signal strength $s^2 = \|\beta_0\|_2^2$ increases linearly with the model size p as shown:

$$s^2 = s^2(\kappa) = r^2 \cdot (\kappa/\zeta), \quad \kappa \in (0, \zeta]. \quad (12)$$

for fixed r^2 and $\zeta > 0$. This models a setting where all coefficients of the regressor η_0 have equal contribution.

Polynomial model: In contrast with the linear model, where the signal strength increases linearly by adding more features during training, the second model captures diminishing returns:

$$s^2 = s^2(\kappa) = r^2 \cdot (1 - (1 + \kappa)^{-\alpha}), \quad \kappa > 0, \quad (13)$$

for some $\alpha \geq 1$. The signal strength s increases with κ , but the increments reduce in size with κ , as decided by α .

B. Risk curves

Figures 1 and 2 assume the polynomial feature model (13) for $\alpha = 2$ and three values of r . Figures 3 and 4 show results with the linear feature model (12) for $\zeta = 5$ and three values of r . Simulation results are shown with crosses ('x'),

²Note that both convex minimization programs have closed form solutions in terms of the Moore-Penrose pseudoinverse $(\mathbf{W}^T \mathbf{W})^\dagger$, i.e., $\hat{\beta} = (\mathbf{W}^T \mathbf{W})^\dagger \mathbf{W}^T \mathbf{y}$. Thus, it is also possible to use random matrix theory tools to analyze the classification performance of $\hat{\beta}$. We find the CGMT more direct to implement; it also provides a unified treatment with the results in [5].

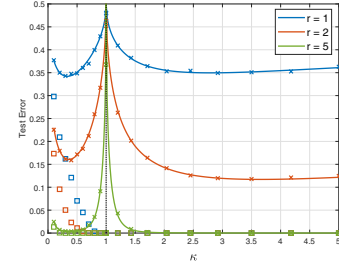


Fig. 1: Test error as a function of the overparametrization ratio κ under a polynomial feature selection rule (with $\alpha = 2$), and with GM data model

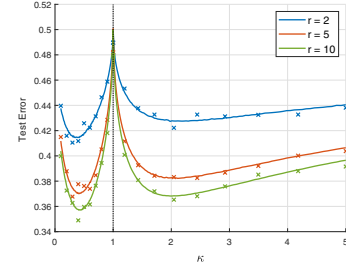


Fig. 2: Test error as a function of the overparametrization ratio κ under a polynomial feature selection rule (with $\alpha = 2$), and with Logistic data model

and theoretical predictions with solid curves. Square-markers show training error, however these have been omitted in few plots to display test-error behavior more clearly. The dotted vertical lines depict the location of the linear interpolation threshold ($\kappa = 1$) according to III.1. Simulation results shown here are for $n = 500$ and the errors are averaged over 300 independent problem realizations. Figure 5 shows a comparison between the predicted performance of the two algorithms, namely GD on square-loss and GD on logistic-loss, with logistic data model. Finally, Figure 6 demonstrates the predicted test error as a function of $\frac{\kappa}{\zeta}$ for three values of ζ . Overall, the simulations validate the theoretical predictions of Propositions III.2 and III.3. Experimental training error behavior also matches with the result of Proposition III.1. The training error is zero with high probability when the training data can be linearly interpolated using the trained regressor under either of the two loss functions. This happens in the regime when κ , our measure of the model complexity exceeds the phase-transition threshold of $\kappa = 1$.

C. Discussion on double descent

The double-descent phenomenon displayed by test error as a function of the overparametrization ratio can be observed from the figures shown. Test error performance has a local minimum in the underparametrized regime. Under the polynomial feature selection model, test error can also attain a local minimum in the overparametrized region for $\kappa < \zeta$. The regime to which the global minimum belongs depends on the data. In linear feature selection model, the curve monotonically decreases in

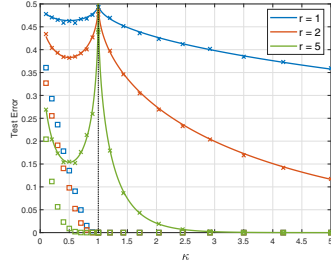


Fig. 3: Test error as a function of κ for the linear feature selection model (cf. (12)) under GM data model.

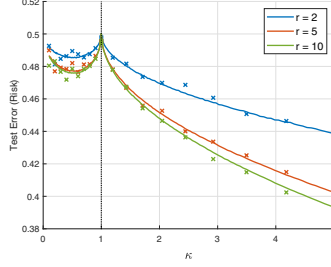


Fig. 4: Test error as a function of κ for the linear feature selection model (cf. (12)) under the Logistic data model.

the region $\kappa \in (1, \zeta]$, thus attaining a local minimum at $\kappa = \zeta$. *On the threshold of phase-transition:* Recall from Proposition III.1 that, for LS, the threshold distinguishes between two regimes: one where data can be linearly interpolated with a regressor of size $p = \kappa n$, and one where it can not. For the logistic loss, the distinction amounts to whether the data is linearly separable or not. In both cases, for model size p larger than what the threshold determines, the empirical loss $\widehat{\mathcal{R}}_{\text{emp}}$ over the training set is zero. This threshold distinguishes between the two U-shapes in the test-error curve. Our results show that this threshold depends in general on both the data (through the models of data – logistic or GM – and through the SNR) and on the training algorithm, which in our setting is determined by the choice of loss function in (4).

Comparison with logistic loss: As observed from Figure 5, the test error of GD on square-loss as well as on logistic-loss exhibit double-descent. However, the central peak (cusp of the curves) depends on the algorithm, i.e. the loss function being minimized. In case of least-squares, it is always 1, whereas for logistic loss $\kappa_{*,\text{LOG}}$ is data-dependent. For $\kappa < \kappa_{*,\text{LOG}}$ the test error of LS and logistic are almost indistinguishable (see [28]). When $\kappa > 1$ the logistic loss is always better, however, the overall minimum, depends on the data SNR. For instance, for $\alpha = 2$ the minimum of the curve corresponding to Logistic loss is below that of the LS curve, but for $\alpha = 5$ the opposite is true. These observations dictate that while logistic loss minimization may appear natural for reducing test error, it is not the best in general. Finally, it is worth observing that the test error of logistic loss is less sensitive to the model size

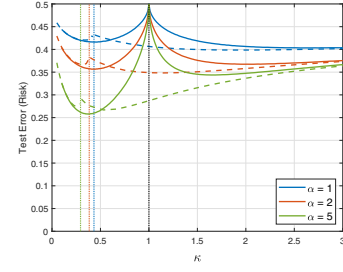


Fig. 5: Test error as a function of κ for the polynomial feature selection model (cf. (13)) under logistic data. Risk curves for square-loss are shown as solid lines and those for logistic-loss are shown as dashed lines.

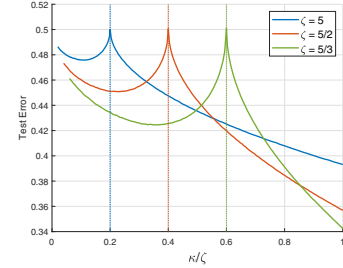


Fig. 6: Test error as a function of $\frac{\kappa}{\epsilon}$ for the linear feature selection model (cf. (12)) under the Logistic data model.

p compared to the LS curve which undergoes a sharp increase close to the threshold $\kappa = 1$.

On the size of the training set: From the theoretical results on the linear interpolation threshold, we notice that increasing the number of training samples increases the interpolation threshold for Logistic loss, but not for LS. This could affect the test error performance accordingly. Also, due to the possibly two local minima, one can infer that more training examples do not necessarily help, and could potentially hurt the generalization performance, in both the underparametrized as well as the overparametrized regimes. One of the observations of [2] is that when the training size grows larger, the interpolation threshold in terms of model complexity "shifts to the right". A similar conclusion can be made from Figure 6. The three curves correspond to three values of n for fixed d , in the linear feature model, with logistic data and square-loss function. One can see that, as n increases, the threshold location, in terms of $\frac{\kappa}{\epsilon}$ increases.

V. FUTURE WORK

In this and the companion paper [5], we study double-descent curves in binary linear classification under simple models with Gaussian features. The proposed setting is simple enough that it allows a principled analytic study of several important features of DD phenomena. We believe that this line of work can be extended in several aspects including the following: better understanding the effect of correlation on the DD curve; similar studies on the effect of label-noise and data-augmentation; and extensions to the multi-class setting.

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